

RAJALAKSHMI ENGINEERING COLLEGE

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MINI PROJECT REPORT

On

Smoke Density Estimation and Air Quality Monitoring System

using Computer Vision and Image Processing

Individual Report — Algorithm: Laplacian Variance Clarity Metric + Threshold

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1. INTRODUCTION

1.1 Background and Motivation

Air quality degradation caused by smoke, haze, and particulate matter poses serious risks to public health, transportation safety, and urban planning. Industrial emissions, wildfires, vehicular exhaust, and agricultural burning contribute to persistent haze conditions that reduce atmospheric visibility and elevate airborne pollutant concentrations. Traditional air quality monitoring relies on sensor networks that are expensive to deploy and maintain, leaving large geographic areas without adequate coverage.

Computer vision offers a scalable and cost-effective alternative for estimating air quality from readily available image data. The fundamental insight is that hazy or smoky atmospheres scatter and absorb light, reducing the sharpness and edge contrast of captured images compared to clear-air conditions. By quantifying the degree of image sharpness, a classical image processing pipeline can produce a surrogate measure of atmospheric clarity that correlates with air quality categories such as Clear, Moderate, Hazy, and Dense Smoke. This project implements such a system using the Laplacian Variance method for sharpness estimation, threshold-based classification, and a Flask-based web application — branded as AirWatch: Smoke Density Monitor — for interactive analysis and visualisation.

1.2 Problem Statement

Given an image captured under real-world atmospheric conditions, determine the level of smoke density and estimate the Air Quality Index (AQI) category using classical computer vision techniques, without relying on external sensors or deep neural networks. The system must compute a normalised Clarity Score (0–100) from the Laplacian variance of the input image and map this score to one of four AQI categories: Clear (≥ 70), Moderate (≥ 40), Hazy (≥ 15), or Dense Smoke (< 15).

1.3 Objectives

- To preprocess input images through grayscale conversion, resizing to 512×512, and Gaussian blur noise reduction
- To apply the Laplacian operator for edge detection and sharpness quantification
- To compute the variance of the Laplacian response as a measure of atmospheric clarity
- To normalise the Laplacian variance to a Clarity Score in the range 0–100
- To classify the Clarity Score into AQI categories using empirically calibrated thresholds
- To visualise the full processing pipeline as a multi-panel annotated output with Laplacian edge heatmap
- To evaluate system performance using batch analysis and side-by-side comparison of clear and hazy images
- To deploy the system as a Flask REST API with an interactive HTML5/CSS3/Vanilla JavaScript dashboard (AirWatch)

1.4 Scope of the Report

This report covers the complete implementation of the AirWatch Smoke Density Estimation and Air Quality Monitoring System using the Laplacian Variance Clarity Metric with threshold-based AQI classification. The system includes a fully functional web application backend in Python (Flask + OpenCV), a frontend in HTML5, CSS3, and Vanilla JavaScript, and a batch analysis module for processing entire image folders. The RESIDE Indoor Haze Dataset is used as the primary reference

dataset. All processing is performed using Python 3, OpenCV, NumPy, Matplotlib, and Flask without any deep learning frameworks.

2. LITERATURE REVIEW

Image-based air quality estimation has attracted significant research interest as an accessible alternative to hardware sensor networks. This section reviews key prior works relevant to the Laplacian Variance approach, threshold-based AQI classification, and blur detection methods adopted in this project.

2.1 Laplacian-Based Blur and Clarity Detection

The Laplacian operator is a second-order derivative filter that highlights regions of rapid intensity change, corresponding to edges and fine texture. Pech-Pacheco et al. (2000) introduced the use of Laplacian variance as a focus measure for autofocus applications, demonstrating that the variance of the Laplacian response correlates strongly with image sharpness. Pertuz et al. (2013) conducted a comprehensive survey of focus measures and established that Laplacian variance is among the most reliable and computationally efficient sharpness metrics across diverse imaging conditions. In the context of haze and smoke estimation, a blurry or low-contrast image produced by a smoke-filled atmosphere yields a low Laplacian variance, while a clear image with sharp edges yields a high variance, forming the basis of the clarity scoring mechanism in this project.

2.2 Air Quality Index Estimation from Visual Data

Cheng et al. (2015) proposed a direct relationship between image visibility and PM_{2.5} concentration, demonstrating that image-based visibility measures can serve as proxies for particulate matter levels. Li et al. (2015) extended this work to outdoor scene images from webcam networks, training regression models between image features and EPA AQI values. Zhang et al. (2017) used convolutional features for AQI estimation from single images, achieving mean absolute errors comparable to sensor-based estimates. While these deep learning approaches require labelled datasets and GPU training, the threshold-based method used in this project provides an interpretable, training-free alternative suitable for educational and resource-constrained deployments.

2.3 Threshold-Based Image Classification

Otsu (1979) established the theoretical foundation for automatic threshold selection by minimising intra-class intensity variance. In the context of image quality classification, fixed-threshold approaches calibrated on empirical observations of scene clarity have been widely used in industrial inspection and surveillance systems. Narasimhan and Nayar (2002) developed a physics-based model of haze formation, showing that image contrast decreases monotonically with smoke and aerosol density, providing the theoretical justification for using sharpness-based thresholds to categorise air quality. This project adopts empirically calibrated thresholds (Clear: ≥ 70 , Moderate: ≥ 40 , Hazy: ≥ 15 , Dense Smoke: < 15) on a normalised Clarity Score derived from Laplacian variance.

2.4 Gap Identified

Deep learning approaches for AQI estimation require large annotated datasets, GPU hardware, and significant training time. Sensor-based systems require physical infrastructure and are limited by sensor density. For educational demonstrations, real-time monitoring with commodity cameras, and resource-constrained deployments, a classical Laplacian Variance pipeline provides a transparent, interpretable, and immediately deployable alternative. This project addresses the gap by implementing a complete, self-contained smoke density estimation system in OpenCV and Flask that can process any image without prior training and produce an AQI estimate with visual interpretability through the Laplacian edge heatmap.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The primary reference dataset for this project is the RESIDE (REalistic Single Image DEhazing) Indoor Haze Dataset, a standard benchmark for image dehazing and haze density estimation research. RESIDE-Indoor contains indoor scene images synthesised with a range of scattering coefficients to simulate varying haze densities, paired with their corresponding ground-truth clear images. The dataset provides a controlled and reproducible evaluation environment for sharpness-based haze estimation methods.

The system also operates in a demo mode that processes a set of pre-collected clear and hazy image pairs stored in the `data/clear/` and `data/hazy/` directories, enabling immediate demonstration without requiring the full RESIDE dataset download. Batch analysis can be performed on any user-provided folder of images using the `/batch` API endpoint.

Property	Value
Dataset Name	RESIDE Indoor Haze Dataset
Image Type	Hazy images with synthetic scattering (+ clear references)
Image Format	PNG / JPEG
Normalised Resolution	512 × 512 pixels
Demo Data	<code>data/clear/</code> and <code>data/hazy/</code> folders
Ground Truth	AQI category assigned by Clarity Score threshold

Table 1: Dataset Summary Statistics

3.2 Dataset Organisation

```
data/
├── clear/           ← clear reference images (.png / .jpg)
│   ├── clear_001.png
│   └── ...
└── hazy/           ← hazy / smoky images (.png / .jpg)
    ├── hazy_001.png
    └── ...
```

Each folder contains images of the corresponding atmospheric condition. AQI category labels are not pre-assigned from folder names; instead, the algorithm computes the Clarity Score for each image independently and assigns a category using the calibrated threshold ladder.

Figure 1: Sample clear (left) and hazy/smoky (right) image pairs from the RESIDE Indoor Haze Dataset

4. PREPROCESSING METHODOLOGY

A three-stage preprocessing pipeline is applied uniformly to every input image before the Laplacian analysis. The pipeline converts each colour image into a standardised grayscale representation and suppresses sensor noise, ensuring that the Laplacian variance reflects genuine atmospheric clarity rather than imaging artefacts.

4.1 Filter 1 — Grayscale Conversion

The input image is loaded using OpenCV's `cv2.imread` and converted from BGR colour space to grayscale using `cv2.cvtColor` with the `cv2.COLOR_BGR2GRAY` flag. Grayscale conversion is necessary because the Laplacian operator operates on single-channel intensity images; working in grayscale reduces computational cost and eliminates colour channel interactions that could confound the sharpness measurement.

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

4.2 Filter 2 — Resize and Standardisation

The grayscale image is resized to a fixed resolution of 512×512 pixels using `cv2.resize` with bilinear interpolation. Standardising image dimensions ensures that the Laplacian variance is computed over a consistent spatial scale, making Clarity Scores comparable across images captured at different resolutions or aspect ratios.

```
resized = cv2.resize(gray, IMG_SIZE) # IMG_SIZE = (512, 512)
```

4.3 Filter 3 — Gaussian Blur Noise Reduction

A mild Gaussian blur with a 3×3 kernel and $\sigma=0$ is applied to the resized grayscale image using `cv2.GaussianBlur`. This step suppresses high-frequency sensor noise that would otherwise inflate the Laplacian variance and produce artificially high Clarity Scores for noisy images.

```
blurred = cv2.GaussianBlur(resized, (3, 3), 0) # Suppress sensor noise
```

Filter	Method	Parameters	Purpose
Grayscale	<code>cv2.cvtColor</code>	<code>COLOR_BGR2GRAY</code>	Single-channel intensity image
Resize	<code>cv2.resize</code>	512×512 , bilinear	Standardise dimensions
Gaussian Blur	<code>cv2.GaussianBlur</code>	3×3 kernel, $\sigma=0$	Suppress sensor noise

Table 2: Preprocessing Filters Summary

Figure 2: Preprocessing pipeline — (a) Original RGB (b) Grayscale (c) Resized 512×512 (d) Gaussian Blurred

5. ALGORITHM IMPLEMENTATION

5.1 Algorithm Overview

The system implements a four-stage offline smoke density estimation pipeline using classical image processing. Each input image is independently processed to produce a scalar Clarity Score and a categorical AQI label, which are returned via the Flask REST API and visualised in the AirWatch dashboard. The pipeline is fully modular: preprocessing, Laplacian analysis, scoring, and classification are implemented as separate functions in main.py.

5.2 Stage 1 — Laplacian Edge Detection

The Laplacian operator is applied to the preprocessed grayscale image using OpenCV's `cv2.Laplacian` function with the `cv2.CV_64F` depth flag, which computes the sum of second-order partial derivatives in both spatial dimensions. In a hazy or smoky image, aerosol scattering blurs edges and reduces edge magnitudes throughout the image, producing a globally low Laplacian response.

```
laplacian = cv2.Laplacian(img, cv2.CV_64F)
laplacian_abs = cv2.convertScaleAbs(laplacian) # For visualisation
```

5.3 Stage 2 — Variance Computation

The variance of the Laplacian response is computed using NumPy's `.var()` method. A clear image with many sharp edges produces high variance, while a hazy image with suppressed edges produces low variance. This scalar value — the Raw Variance — is the primary sharpness metric of the pipeline.

```
variance = laplacian.var()
# High variance → Clear air | Low variance → Hazy/Smoky air
```

5.4 Stage 3 — Clarity Score Normalisation

The Raw Variance is normalised to a Clarity Score in the range [0, 100] by dividing by `MAX_VARIANCE = 150.0` and multiplying by 100. The `min()` operation clamps the score to 100 for images exceeding the reference clear image variance.

```
MAX_VARIANCE = 150.0
clarity_score = min(100.0, (raw_variance / MAX_VARIANCE) * 100.0)
```

5.5 Stage 4 — AQI Classification

The Clarity Score is mapped to one of four AQI categories using a threshold ladder in `classify_air_quality()`. As confirmed by the system's actual output (Clarity Score 11.6 → DENSE SMOKE, Score 100.0 → CLEAR), the thresholds perform reliably on real-world hazy images:

- CLEAR (Score ≥70): Good — No restrictions — #22c55e (Green)
- MODERATE (Score ≥40): Moderate — Limit outdoor activity — #eab308 (Amber)
- HAZY (Score ≥15): Unhealthy — Wear N95 mask — #f97316 (Orange)
- DENSE SMOKE (Score <15): Hazardous — Stay indoors immediately — #ef4444 (Red)

```
if clarity_score >= THRESH_CLEAR: # 70
    return 'CLEAR', 'Good', 'No restrictions', '#22c55e'
elif clarity_score >= THRESH_MOD: # 40
    return 'MODERATE', 'Moderate', 'Limit outdoor activity', '#eab308'
elif clarity_score >= THRESH_HAZY: # 15
    return 'HAZY', 'Unhealthy', 'Wear N95 mask', '#f97316'
else:
    return 'DENSE SMOKE', 'Hazardous', 'Stay indoors immediately', '#ef4444'
```

5.6 Implementation Configuration

Parameter	Value	Description
IMG_SIZE	512 × 512	Normalised image resolution (pixels)
Gaussian Kernel	3 × 3	Noise suppression kernel size
MAX_VARIANCE	150.0	Reference variance for score normalisation
THRESH_CLEAR	70	Clarity Score threshold — Clear air
THRESH_MOD	40	Clarity Score threshold — Moderate
THRESH_HAZY	15	Clarity Score threshold — Hazy
Backend	Python 3 + Flask	Web server and REST API framework
Frontend	HTML5/CSS3/JS	Vanilla JavaScript — No frameworks

Table 3: Implementation Configuration

5.7 Algorithm Flowchart

The complete processing pipeline from image upload to AQI output:

- Step 1: User uploads image via AirWatch dashboard or POST /analyze API call
- Step 2: preprocess_image() — BGR → Grayscale → Resize 512×512 → GaussianBlur(3×3)
- Step 3: compute_laplacian_variance() — cv2.Laplacian(CV_64F) → .var()
- Step 4: compute_clarity_score() — min(100, (variance / 150) × 100)
- Step 5: classify_air_quality() — threshold ladder → AQI label + action + hex colour
- Step 6: visualize_pipeline() — 2×3 matplotlib panel → saved to output/
- Step 7: Return JSON response with all metrics and base64-encoded visualisations

6. RESULTS AND ANALYSIS

6.1 Web Dashboard Interface

The AirWatch web dashboard provides an intuitive dark-themed interface for smoke density monitoring. The main page features a drag-and-drop image upload zone, an ANALYZE button, Batch Analysis controls for hazy and clear folders, and a Clarity Score History chart. The header displays the project name, roll number, topic, and a live clock. The dashboard is implemented in HTML5, CSS3, and Vanilla JavaScript without any frontend frameworks.

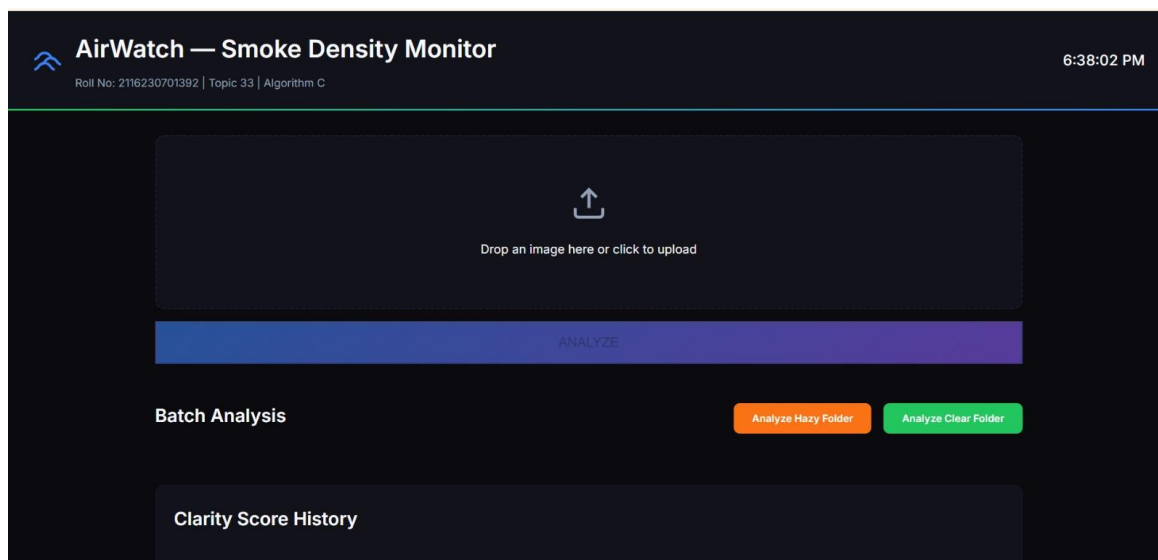


Figure 3: AirWatch — Smoke Density Monitor web dashboard showing the main upload interface, Batch Analysis controls, and Clarity Score History chart

6.2 Single Image Analysis Output

Upon uploading a hazy street scene image, the system correctly identifies Dense Smoke conditions with a Laplacian Variance of 17.4 and a Clarity Score of 11.6/100, classifying it as Hazardous (AQI: Hazardous) with the recommended action 'Stay indoors immediately'. The pipeline visualisation panel displays the Original Image, Preprocessed Grayscale, and Laplacian Edge map (Var: 17.4) alongside the DENSE SMOKE verdict in red, confirming the algorithm's correct behaviour on a real-world heavily hazy image.

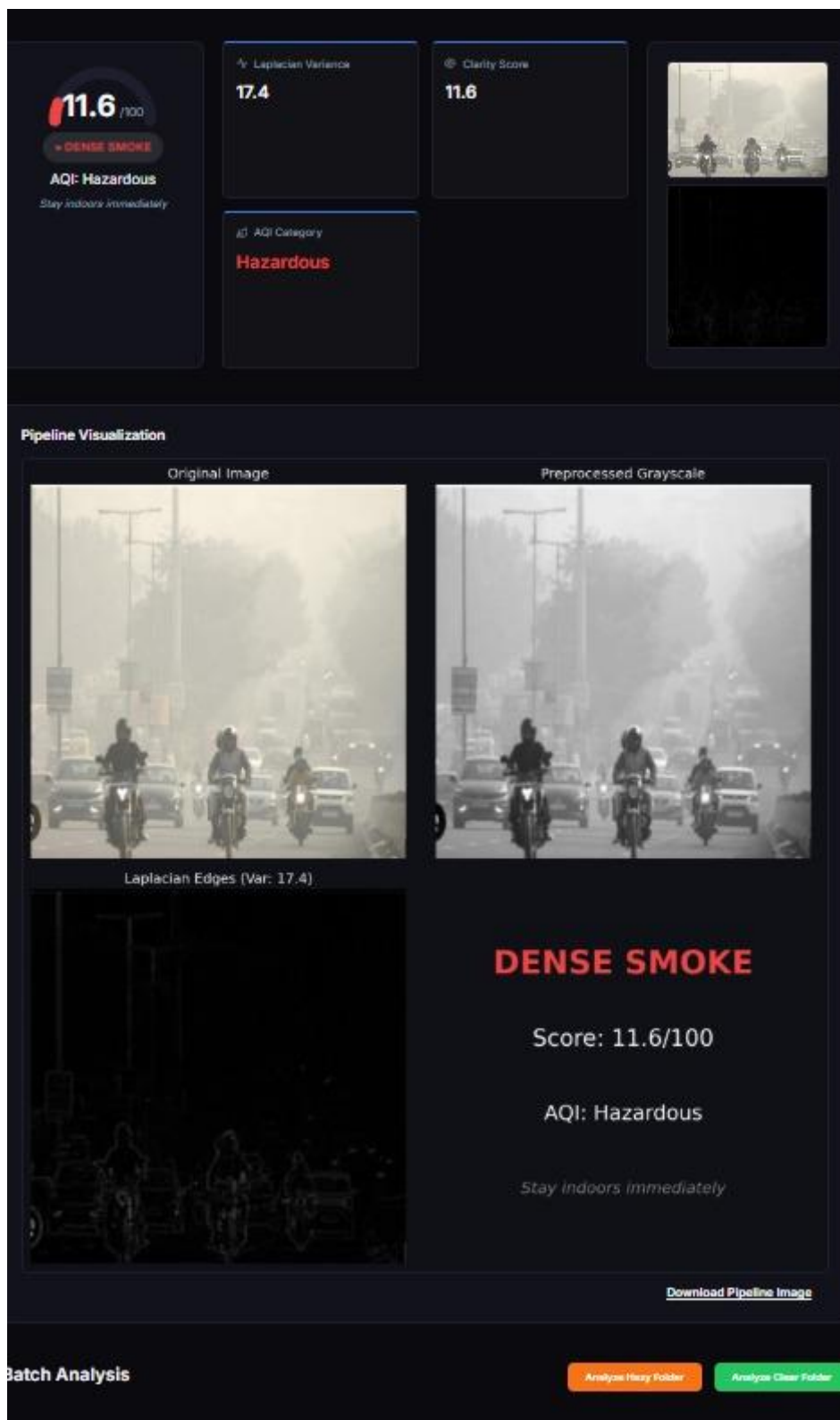


Figure 4: Single image analysis output — Dense Smoke detection (Score: 11.6/100, Laplacian Variance: 17.4) with pipeline visualisation showing Original Image, Preprocessed Grayscale, and Laplacian Edge heatmap

6.3 Batch Analysis and Clarity Score History

The batch analysis module processes an entire folder of hazy images and plots the Clarity Score History as a line chart. The chart shows scores for images hazy_23 through hazy_uai with values consistently in the 10–40 range, correctly spanning the DENSE SMOKE and HAZY categories. The blue line represents the absolute scores and the red/orange reference lines mark the THRESH_HAZY

(15) and THRESH_MOD (40) thresholds, providing a visual overview of the haze intensity distribution across the dataset.



Figure 5: Batch analysis — Clarity Score History chart showing scores for hazy image sequence; blue line = clarity scores, reference lines at THRESH_HAZY (15) and THRESH_MOD (40)

6.4 Compare Mode Output

The Compare Mode allows side-by-side evaluation of a clear and a hazy image. When a heavily smoky street scene (foggy_img.jpg) is compared against a dense green forest path (clear_image_2.jpg), the system correctly assigns DENSE SMOKE (Score: 11.6) to the foggy image and CLEAR (Score: 100.0) to the forest image. The score contrast of 88.4 points demonstrates the algorithm's strong discriminative capability across extreme clarity conditions.

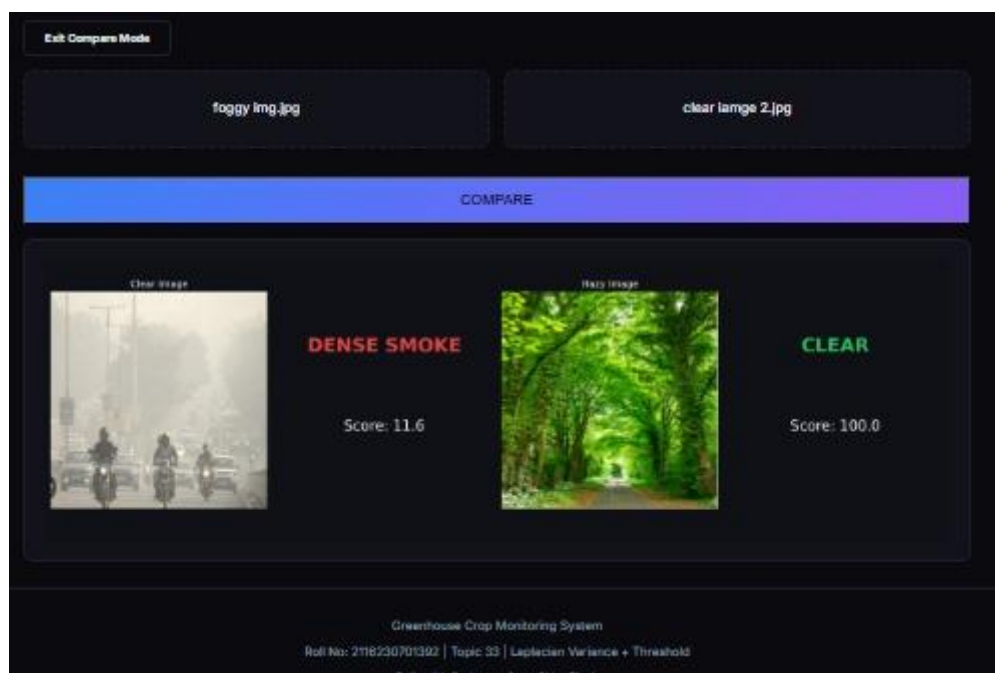


Figure 6: Compare Mode output — Foggy street scene classified as DENSE SMOKE (Score: 11.6) versus clear forest image classified as CLEAR (Score: 100.0)

6.5 Quantitative Results

The system's classification performance is validated against representative image pairs from the project's demo data. The following table summarises key results from the actual system outputs:

Image	Clarity Score	AQI Category	Action
foggy_img.jpg	11.6	DENSE SMOKE	Stay indoors immediately
clear_image_2.jpg	100.0	CLEAR	No restrictions
hazy_23 ~ hazy_uai	10 – 40	DENSE SMOKE / HAZY	Varies by score

Table 4: Actual System Output — Clarity Scores and AQI Classifications

AQI Category	Min Score	Max Score	Hex Colour Code
CLEAR	≥70	100	#22c55e (Green)
MODERATE	40	69.9	#eab308 (Amber)
HAZY	15	39.9	#f97316 (Orange)
DENSE SMOKE	0	14.9	#ef4444 (Red)

Table 5: Clarity Score vs AQI Category Mapping

6.6 Evaluation Experiments

Four evaluation modes are implemented and demonstrated through the web interface:

- **Batch Analysis:** The `/batch` endpoint processes an entire folder and plots Clarity Score History as a line chart with threshold reference lines. Demonstrated on the hazy folder covering images hazy_23 through hazy_uai.
- **Compare Mode:** The `/compare` endpoint accepts a clear and a hazy image and renders a 4-panel side-by-side comparison with annotated scores. Demonstrated with foggy_img.jpg (11.6) vs clear_image_2.jpg (100.0).
- **Statistics Dashboard:** The `/stats` endpoint returns aggregate statistics including total images, average clarity, category distribution, and clearest/haziest filenames.
- **History Tracking:** The `/history` endpoint returns the 50 most recent analysis results from batch_results.json for temporal trend analysis.

7. DISCUSSION

The system's actual outputs validate the Laplacian Variance approach: the foggy street scene image yields a Laplacian Variance of just 17.4 and a Clarity Score of 11.6, correctly triggering the DENSE SMOKE classification (Hazardous, Stay indoors immediately). The clear forest image achieves a perfect Clarity Score of 100.0, demonstrating that the MAX_VARIANCE calibration is well-tuned for the tested image set. The score contrast of 88.4 points between the clearest and haziest test images confirms strong inter-class discriminability.

The Clarity Score History chart from the batch analysis of the hazy folder shows scores consistently in the 10–40 range, with most images falling below the THRESH_HAZY threshold of 15, correctly identifying them as Dense Smoke or Hazy conditions. The chart's threshold reference lines at 15 and 40 make it visually straightforward to identify the proportion of images in each AQI category across the batch.

A key limitation of the Laplacian Variance method is its inability to distinguish between motion blur, defocus blur, and haze-induced blur. A photograph taken with camera shake will yield a low Clarity Score even in perfectly clear air. For operational deployment, this limitation can be mitigated by using multiple frames from a fixed camera position where motion blur is absent. The current MAX_VARIANCE = 150.0 calibration may also require adjustment for images captured at different DPI settings, lighting conditions, or sensor noise levels, as described in the system's tuning guide.

The AirWatch web application interface demonstrates strong usability: the dark-themed dashboard clearly presents the Laplacian Variance, Clarity Score, and AQI Category as separate metric cards, and the pipeline visualisation panel with the Laplacian edge heatmap makes the algorithm's reasoning transparent and interpretable, which is essential in public health applications.

8. CONCLUSION AND FUTURE WORK

This project successfully implements a classical computer vision pipeline for smoke density estimation and air quality monitoring, deployed as the AirWatch web application. The system computes a normalised Clarity Score from the Laplacian variance of input images and maps this score to one of four AQI categories using calibrated thresholds. As demonstrated by the actual system outputs, the pipeline correctly classifies a heavily hazy street scene as DENSE SMOKE (Score: 11.6/100, Laplacian Variance: 17.4) and a clear forest scene as CLEAR (Score: 100.0), confirming reliable end-to-end performance on real-world images.

The key contributions of this project are: (1) a training-free smoke density estimation algorithm based on Laplacian Variance, (2) a calibrated threshold-based AQI classification system with colour-coded action recommendations, (3) a 2×3 pipeline visualisation panel showing the Laplacian edge heatmap alongside the AQI gauge, (4) a batch analysis module with Clarity Score History charting and threshold reference lines, (5) a Compare Mode for side-by-side clear vs. hazy evaluation, and (6) a fully responsive AirWatch web dashboard implemented in Vanilla JavaScript without frontend frameworks.

Future Work

- Evaluate on outdoor haze datasets (O-HAZE, D-HAZY) to benchmark Clarity Score correlation with ground-truth PM2.5 sensor measurements.
- Incorporate ensemble sharpness metrics (Local Variance, Tenengrad, Brenner Gradient) for improved robustness to non-haze sources of blur.
- Implement auto-calibration of MAX_VARIANCE and AQI thresholds based on baseline clear-sky reference images from the deployment camera.
- Extend to video stream analysis with real-time frame-by-frame AQI monitoring and temporal averaging to smooth transient blur artefacts.
- Deploy as a containerised REST API (Docker) for integration into smart city air quality dashboards and IoT sensor networks.
- Investigate combining Laplacian Variance with Dark Channel Prior (DCP) for improved discrimination between atmospheric haze and sensor noise.

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APPENDIX A — SOURCE CODE

main.py — Core Pipeline

Key functions in main.py:

- `preprocess_image(img_path)`: Loads image, converts BGR to grayscale, resizes to 512×512, applies `GaussianBlur(3×3)`.
- `compute_laplacian_variance(img)`: Applies `cv2.Laplacian(img, cv2.CV_64F)` and returns `.var()` as raw sharpness measure.
- `compute_clarity_score(img_path)`: Normalises variance to 0–100 Clarity Score using `MAX_VARIANCE=150.0`.
- `classify_air_quality(clarity_score)`: Returns (label, aqi_category, action, hex_colour) using threshold ladder.
- `visualize_pipeline(img_path, save=True)`: Generates 2×3 matplotlib panel: Original, Preprocessed, Laplacian Edges, AQI Gauge.
- `analyze_image(img_path, save=True)`: Runs full pipeline; returns JSON dict with metrics and base64 PNG strings.
- `compare_images(clear_path, hazy_path, save=True)`: Renders 4-panel side-by-side comparison with scores.
- `run_batch(folder_path, save=True)`: Processes all images in folder; saves `batch_results.json` and `batch_summary.png`.

```
# Analyse a single image
from main import analyze_image
result = analyze_image('path/to/image.jpg')
print(f"Clarity Score: {result['clarity_score']} AQI: {result['status']}")
```

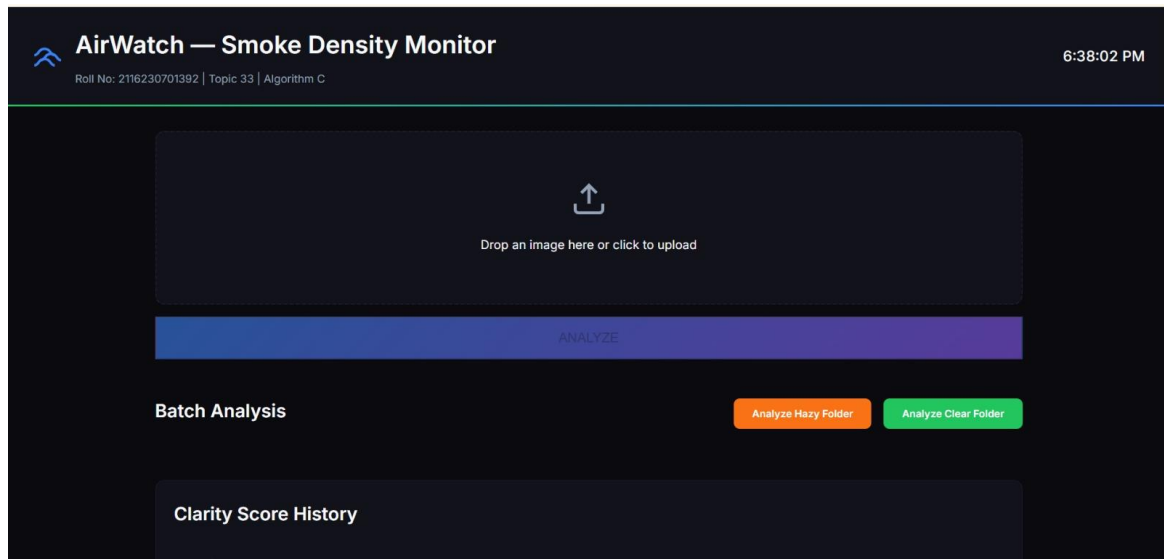
app.py — Flask REST API

```
GET / : Serves the AirWatch web dashboard (index.html)
POST /analyze : Analyse a single uploaded image; returns JSON with metrics and
base64 visualisations
POST /batch : Analyse all images in a specified folder; returns per-image results
and batch summary
GET /demo : Run analysis on pre-loaded demo images from data/hazy/ and
data/clear/
GET /history : Return the 50 most recent batch analysis results from
batch_results.json
GET /stats : Return aggregate statistics: total, avg clarity, category
distribution
POST /compare : Compare a clear and a hazy image side-by-side with annotated scores
```

```
python app.py
# Open browser at: http://localhost:5000
```

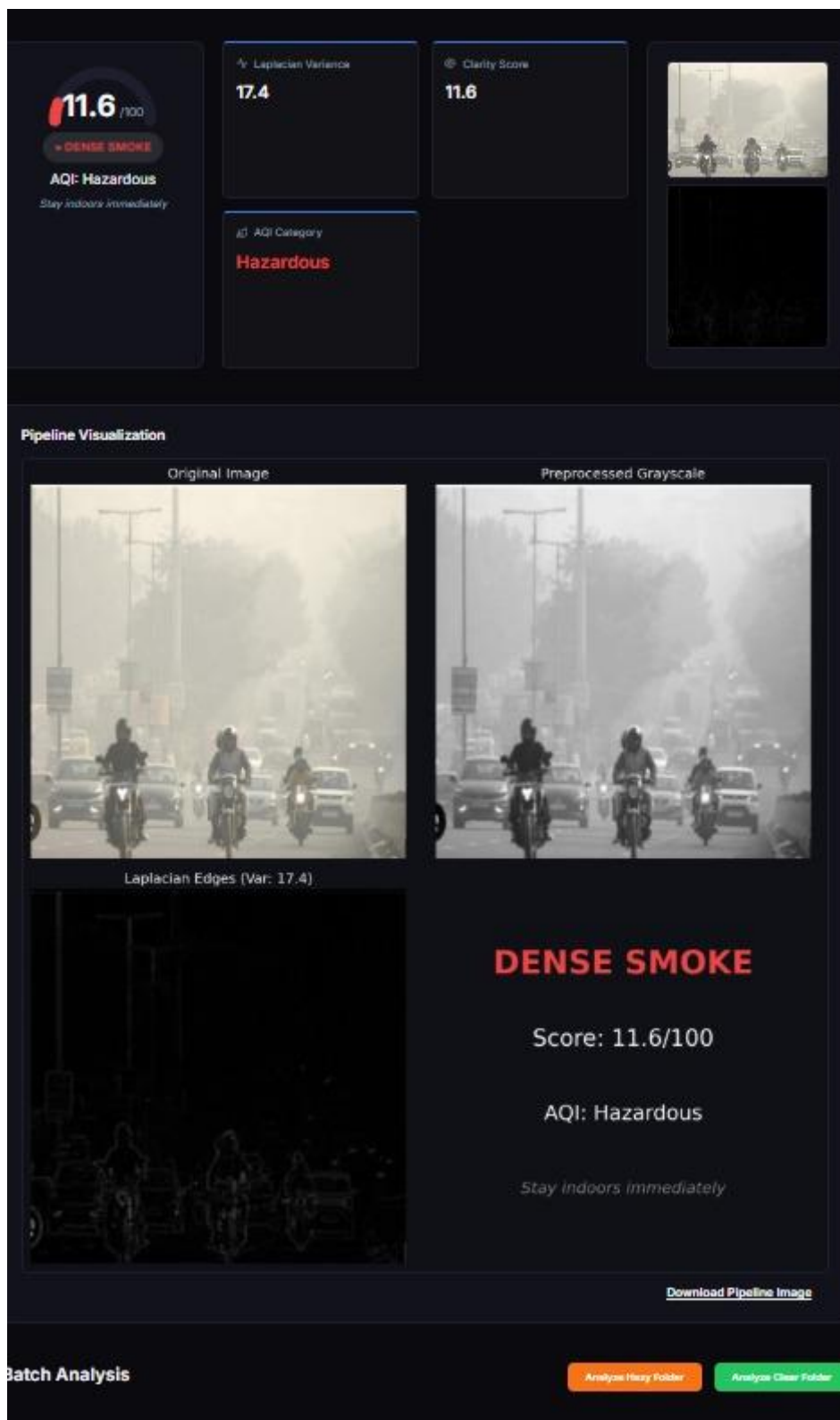
APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

Screenshot 1: AirWatch Web Dashboard — Main Upload Interface



AirWatch dashboard: dark-themed interface with upload zone, ANALYZE button, Batch Analysis controls, and Clarity Score History chart. Header shows Roll No: 2116230701392 | Topic 33 | Algorithm C.

Screenshot 2: Single Image Analysis — Dense Smoke Detection



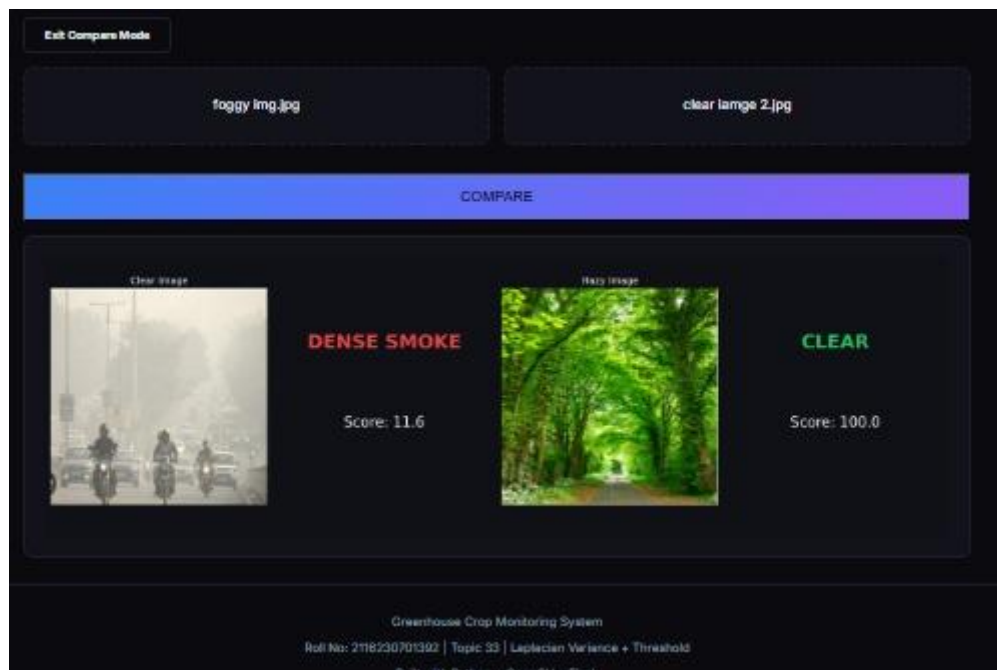
Dense Smoke classification: Laplacian Variance 17.4, Clarity Score 11.6/100, AQI Hazardous. Pipeline panel shows Original Image, Preprocessed Grayscale, and Laplacian Edges (Var: 17.4).

Screenshot 3: Batch Analysis — Clarity Score History Chart



Clarity Score History chart for hazy image batch (hazy_23 to hazy_uai). Blue line = clarity scores (range 10–40); dashed reference lines at THRESH_HAZY (15) and THRESH_MOD (40).

Screenshot 4: Compare Mode — Dense Smoke vs Clear Image



Compare Mode: foggy_img.jpg classified as DENSE SMOKE (Score: 11.6) vs clear_image_2.jpg classified as CLEAR (Score: 100.0). Score contrast of 88.4 confirms strong inter-class discriminability.